THE MONOMIALS OF DEGREE AT MOST N FORM A BASIS OF THE N-TH FILTRATION LAYER OF THE UNIVERSAL ASKEY-WILSON ALGEBRA

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source sources/sigma-2011-069-askey-wilson/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Three references to other labelled items occur inside the counted spans (Definition def:ui, Lemma lem:almostcom, Theorem thm:basis); each is recorded once as a navigation stub with the matching kind and the author's own number, and every step that consumes them stays inside the counted span.

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Allowed standard prerequisites (editorial).

- Linear independence of the monomials in the universal Askey-Wilson algebra. Same source, Theorem labelled thm:basis; invoked inside the counted proof at lines 1361-1363. Gives the independence half of the statement outright.
- The filtration whose n-th layer is spanned by words of length at most n. Same source, Definition labelled def:ui; invoked inside the counted proof at lines 1365-1376. Supplies the spanning set by words in the generators, including the base case of the induction.
- Swapping two adjacent letters modulo the previous layer. Same source, Lemma labelled lem:almostcom; invoked inside the counted proof at lines 1385-1392. Lets any word be reordered modulo Delta_{n-1}, producing the monomial form needed for the spanning claim.

Source reference. Original definition 1.23: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 1366–1366. Navigation pointer for \ref{def:ui} used inside the counted span (line 1366); the argument that consumes it is retained in the span.

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Source reference. Original lemma 1.25: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 1385–1385. Navigation pointer for \ref{lem:almostcom} used inside the counted span (line 1385); the argument that consumes it is retained in the span.

Source reference. Original theorem 1.21: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 1363–1363. Navigation pointer for \ref{thm:basis} used inside the counted span (line 1363); the argument that consumes it is retained in the span.

Theorem 0.1. *For all integers $n \geq 0$ the following is a basis for the $\mathbb{F}$-vector space Δ_n:*

$$A^i B^j C^k \alpha^r \beta^s \gamma^t, \qquad i,j,k,r,s,t \geq 0, \qquad i+j+k+r+s+t \leq n. \tag{1}$$

Proof. The elements (1) are linearly independent by Theorem 1.21. We show that the elements (1) span Δ_n. We will use induction on n. Assume $n \geq 2$; otherwise the result holds by Definition 1.23. By Definition 1.23 Δ_n is spanned by the set of elements of the form $x_1 x_2 \cdots x_n$ where $x_i \in \{1, A, B, C, \alpha, \beta, \gamma\}$ for $1 \leq i \leq n$. Therefore Δ_n is spanned by the set of elements of the form $x_1 x_2 \cdots x_m$ where $0 \leq m \leq n$ and $x_i \in \{A, B, C, \alpha, \beta, \gamma\}$ for $1 \leq i \leq m$. Therefore Δ_n is spanned by Δ_{n-1} together with the set of elements of the form $x_1 x_2 \cdots x_n$ where $x_i \in \{A, B, C, \alpha, \beta, \gamma\}$ for $1 \leq i \leq n$. Consider such an element $x_1 x_2 \cdots x_n$. By Lemma 1.25 and since each of α, β, γ is central, we find that for $2 \leq j \leq n$,

$$x_1 \cdots x_{j-1} x_j \cdots x_n \in \mathbb{F} x_1 \cdots x_j x_{j-1} \cdots x_n + \Delta_{n-1}.$$

By the above comments Δ_n is spanned by Δ_{n-1} together with the set

$$A^i B^j C^k \alpha^r \beta^s \gamma^t, \qquad i,j,k,r,s,t \geq 0, \qquad i+j+k+r+s+t = n.$$

By this and induction Δ_n is spanned by the elements (1). We have shown that the elements (1) form a basis for the $\mathbb{F}$-vector space Δ_n. □

References

[1] Alperin R.C., Notes: $\mathrm{PSL}_2(\mathbb{Z}) = \mathbb{Z}_2 \star \mathbb{Z}_3$, *Amer. Math. Monthly* **100** (1993), 385–386.